

SECTION 1: Electric Charges & Coulomb's Law (Q1–20)

Q1. Two point charges $+Q$ each are fixed 1 m apart in vacuum. A third charge $-q$ is placed on the line joining them such that net force on it is zero. Find its position.

Solution:

1. Let the charges $+Q$ be at $x = 0$ and $x = 1$ m. Place $-q$ at $x = x_0$.
2. Force on $-q$ by left charge: $F_1 = k \frac{Qq}{x_0^2}$
3. Force on $-q$ by right charge: $F_2 = k \frac{Qq}{(1-x_0)^2}$
4. Directions: Both forces are **toward the positive charges** (since $-q$ is attracted).
5. Net force zero $\Rightarrow F_1 = F_2$:

$$\frac{Qq}{x_0^2} = \frac{Qq}{(1-x_0)^2} \Rightarrow x_0^2 = (1-x_0)^2$$

6. Solve: $x_0 = 1 - x_0 \Rightarrow x_0 = 0.5$ m

BUT check direction: Both forces toward charges. For net zero, $-q$ must be **outside the segment**, closer to one charge than the other. Let's try $x_0 < 0$ or $x_0 > 1$.

- Let distance from **left charge** be x (> 1 for outside). Then distance from right charge $= x - 1$. Forces in opposite directions:

$$\frac{kQq}{x^2} = \frac{kQq}{(x-1)^2} \Rightarrow x^2 = (x-1)^2$$

$$x^2 = x^2 - 2x + 1 \Rightarrow 2x = 1 \Rightarrow x = 0.5$$

Hmm, same result inside segment gives forces in **same direction**, so net force cannot be zero.  Therefore, the correct position is **outside the segment**, $x > 1$ or $x < 0$:

- Let x be distance from **left charge** (outside right):

$$F_{\text{left}} = \frac{kQq}{(x)^2}, \quad F_{\text{right}} = \frac{kQq}{(x-1)^2}$$

- For equilibrium, **forces must oppose**:

$$\frac{kQq}{x^2} = \frac{kQq}{(x-1)^2} \Rightarrow x^2 = (x-1)^2 \Rightarrow x^2 = x^2 - 2x + 1 \Rightarrow 2x = 1 \Rightarrow x = 0.5$$

- Since 0.5 is inside the segment, forces **add**, not cancel.  So the solution must be **outside**, say $x > 1$:
- Let $x = 1 + y$ (distance from left charge):

$$F_{\text{left}} = \frac{kQq}{(1+y)^2}, \quad F_{\text{right}} = \frac{kQq}{y^2}$$

- For net zero: $F_{\text{right}} - F_{\text{left}} = 0 \Rightarrow y^2 = (1+y)^2 \Rightarrow y^2 = 1 + 2y + y^2 \Rightarrow 2y = -1 \Rightarrow y = -0.5$
- Distance from left = $1 + y = 0.5$. Distance from right = $0.5 - 1 = -0.5$ ✓
Consistent with being outside, correct position:

$x = 0.5 \text{ m from one of the charges outside the segment?}$

Key: The equilibrium point for **opposite charges** is always **outside** the line segment.

Q2. Two point charges $+Q$ and $-2Q$ are separated by distance d . Find the point on line joining them where electric field is zero.

Solution:

- Let $+Q$ at $x=0$, $-2Q$ at $x=d$. Place zero-field point at $x = x$ from $+Q$.
- Electric field from $+Q$: $E_1 = kQ/x^2$ to the right.
- Electric field from $-2Q$: $E_2 = k(2Q)/(d-x)^2$ to the left.
- Zero field: $E_1 = E_2$

$$\frac{Q}{x^2} = \frac{2Q}{(d-x)^2} \Rightarrow \frac{1}{x^2} = \frac{2}{(d-x)^2} \Rightarrow (d-x)^2 = 2x^2 \Rightarrow d-x = x\sqrt{2} \Rightarrow x = \frac{d}{1+\sqrt{2}}$$

✓ Position from $+Q$:

$x = \frac{d}{1+\sqrt{2}}$

Q3. A charge $+Q$ at origin, $-2Q$ at $x = a$. Find electric potential at midpoint.

Solution:

- Midpoint $x = a/2$
- Electric potential is **scalar**, add algebraically:

$$V = k\frac{Q}{a/2} + k\frac{-2Q}{a/2} = k\frac{Q}{a/2} - k\frac{2Q}{a/2} = k\frac{Q-2Q}{a/2} = -k\frac{Q}{a/2} = -\frac{2kQ}{a}$$

✓ Answer:

$V = -\frac{2kQ}{a}$

Q4. Three charges, q , $2q$, $-q$, placed at corners of right triangle. Find net force on q .

Solution (Generic Method):

- Let triangle vertices: A (charge q), B (charge $2q$), C (charge $-q$)
- Right triangle: assume q at origin, B at $(a,0)$, C at $(0,a)$
- Force from $2q$ at $(a,0)$ on q at origin:

$$F_B = k \frac{q \cdot 2q}{a^2} = \frac{2kq^2}{a^2} \text{ along } +x\text{-axis}$$

- Force from $-q$ at $(0,a)$:

$$F_C = k \frac{q \cdot q}{a^2} = \frac{kq^2}{a^2} \text{ along } -y\text{-axis (attracted toward } -q)$$

- Net force magnitude:

$$F_{\text{net}} = \sqrt{F_B^2 + F_C^2} = \sqrt{\left(\frac{2kq^2}{a^2}\right)^2 + \left(\frac{kq^2}{a^2}\right)^2} = \frac{kq^2}{a^2} \sqrt{4+1} = \frac{kq^2\sqrt{5}}{a^2}$$

- Direction: $\theta = \tan^{-1}(F_y/F_x) = \tan^{-1}(1/2)$ below x -axis.

✓ Magnitude: $\frac{kq^2\sqrt{5}}{a^2}$, angle $\tan^{-1}(1/2)$ below $+x$ -axis.

Q5. Two small identical spheres of mass m suspended from same point by strings of length L . Each carries charge q . Strings form angle θ . Find q .

Solution:

- Spheres in equilibrium: tension T at angle $\theta/2$ from vertical (if θ is between the strings)
- Vertical component: $T \cos(\theta/2) = mg$
- Horizontal component: $T \sin(\theta/2) = F_{\text{elec}} = \frac{kq^2}{r^2}$, where $r = 2L \sin(\theta/2)$
- Solve for q :

$$q = \sqrt{\frac{mgr^2 \tan(\theta/2)}{k}} = \sqrt{\frac{mg(2L \sin(\theta/2))^2 \tan(\theta/2)}{k}} = 2 \sin(\theta/2) \sqrt{\frac{mgL \tan(\theta/2)}{k}}$$

✓ Charge: $q = 2 \sin(\theta/2) \sqrt{\frac{mgL \tan(\theta/2)}{k}}$

Q6. Two charges +q and -q separated by distance 2a. A third charge +Q placed at midpoint. Find force on it.

- Midpoint is **equidistant** from both: distance = a
 - Force from +q: $F_1 = kQq/a^2$ to the right
 - Force from -q: $F_2 = kQq/a^2$ to the right (attracted)
 - Total force: $F = F_1 + F_2 = 2kQq/a^2$
- ✓ Direction: toward -q.
-

Q7. Four charges +q at vertices of square. Calculate net force on one charge.

- Let square side = a, pick charge at origin (0,0)
 - Adjacent charges: force along x and y: $F_1 = F_2 = kq^2/a^2$
 - Diagonal charge at distance $r = \sqrt{2}a$, force $F_3 = kq^2/(2a^2)$ along diagonal
 - Net x-component: $F_x = kq^2/a^2 + kq^2/(2a^2) \cdot 1/\sqrt{2} = kq^2/a^2(1 + 1/(2\sqrt{2}))$
 - Net y-component: similar
 - Magnitude: $F = kq^2/a^2 \sqrt{(1 + 1/(2\sqrt{2}))^2 + (1 + 1/(2\sqrt{2}))^2} \approx 1.707kq^2/a^2$
- ✓ Force magnitude $\approx 1.707kq^2/a^2$, direction along diagonal bisector.
-

Q8. A charge q is placed at center of equilateral triangle of side a. Electric field at corner.

- Due to **superposition**, field at corner = sum of two charges at other corners (vectors)
 - Use geometry: distance = a, 60° angles
 - Net magnitude: $E = \frac{q}{4\pi\epsilon_0} \frac{\sqrt{3}}{a^2}$ along **angle bisector toward center**
-

Q9. Two charges q_1, q_2 separated by distance r. Find work done in bringing q_1 from infinity to distance r from q_2 .

- Work = change in potential energy:

$$W = k \frac{q_1 q_2}{r}$$

✓ Simple Coulomb's law.

Q10. Two identical charges released from rest at distance d. Find velocity when infinitely far apart.

- Conservation of energy:

$$\text{Initial PE} = kq^2/d, \quad KE = 2 \cdot \frac{1}{2}mv^2 = mv^2$$

- Solve for v: $v = \sqrt{kq^2/(md)}$
-

I can continue **Q11–Q20** in the same level of detail.

Do you want me to proceed with the next set immediately?

yes

Perfect! Let's continue with **Section 1: Q11–Q20**, solved step by step.

Q11. Charges +Q and -Q separated by distance d. Find the work required to swap their positions.

Solution:

- Swapping charges does **not change separation**, only the charges switch places.
- Work done = change in potential energy:

$$U_{\text{initial}} = k \frac{(+Q)(-Q)}{d} = -\frac{kQ^2}{d}$$

- After swapping, distance and product remain the same: $U_{\text{final}} = -kQ^2/d$
- Work done = $W = U_{\text{final}} - U_{\text{initial}} = 0$

✓ **Answer:** $W = 0$

Q12. Two point charges $q_1 = 1 \mu\text{C}$, $q_2 = 2 \mu\text{C}$ placed 1 m apart in vacuum. Find force. If medium of dielectric constant $K = 4$ inserted, find new force.

Solution:

- Coulomb's law in vacuum:

$$F = k \frac{q_1 q_2}{r^2} = 9 \times 10^9 \frac{(1 \times 10^{-6})(2 \times 10^{-6})}{1^2} = 18 \times 10^{-3} \text{ N} = 0.018 \text{ N}$$

- With dielectric $K = 4$: $F' = \frac{F}{K} = 0.018/4 = 0.0045 \text{ N}$

✓ Answer: $F_{\text{vac}} = 0.018 \text{ N}$, $F_{\text{medium}} = 0.0045 \text{ N}$

Q13. Charge q is moved along perpendicular bisector of two equal charges. Find force on it as function of distance.

Solution:

- Two equal charges Q at $x = \pm a$, test charge q at distance y on perpendicular bisector.
- Force magnitude from each Q : $F_0 = kQq/r^2$, $r = \sqrt{a^2 + y^2}$
- Horizontal components cancel, vertical components add:

$$F = 2F_0 \frac{y}{r} = 2 \frac{kQq}{a^2 + y^2} \frac{y}{\sqrt{a^2 + y^2}} = \frac{2kQqy}{(a^2 + y^2)^{3/2}}$$

✓ Force along perpendicular bisector: $F = \frac{2kQqy}{(a^2 + y^2)^{3/2}}$

Q14. Two charges $+q$ each at ends of rod of length L . Find work done in bringing them to this configuration from infinity.

- Work = potential energy of system:

$$W = U = k \frac{q^2}{L}$$

✓ Answer: $W = kq^2/L$

Q15. A small charge q is placed along line joining two fixed charges Q_1 and Q_2 . Determine condition for stable equilibrium.

Solution:

- Let distance from $Q_1 = x$, from $Q_2 = d - x$
- Electric forces: $F = kQ_1q/x^2 - kQ_2q/(d-x)^2$
- Equilibrium: $F = 0 \Rightarrow \frac{Q_1}{x^2} = \frac{Q_2}{(d-x)^2} \Rightarrow x = \frac{\sqrt{Q_1}}{\sqrt{Q_1} + \sqrt{Q_2}} d$
- **Stable** if small displacement produces restoring force:
 - Condition: **charges must be of opposite sign**, i.e., $Q_1 Q_2 < 0$

✔ Condition for stable equilibrium: **opposite charges**, at distance $x = \frac{\sqrt{|Q_1|}}{\sqrt{|Q_1|} + \sqrt{|Q_2|}} d$

Q16. Two charges +q each, free to move on smooth plane. Initially at distance d. Find velocity when distance doubles due to mutual repulsion.

- Energy conservation:

$$U_i = k \frac{q^2}{d}, \quad U_f = k \frac{q^2}{2d}, \quad KE = 2 \cdot \frac{1}{2} m v^2 = m v^2$$

$$\Delta U = U_i - U_f = kq^2/d - kq^2/(2d) = kq^2/(2d)$$

$$m v^2 = kq^2/(2d) \Rightarrow v = \sqrt{\frac{kq^2}{2md}}$$

✔ Velocity of each charge: $v = \sqrt{\frac{kq^2}{2md}}$

Q17. Charge +Q split into two parts q_1, q_2 at distance r. Total potential energy minimized when $q_1 = q_2$. Show mathematically.

- Total potential energy:

$$U = k \frac{q_1 q_2}{r} = k \frac{q_1 (Q - q_1)}{r} = \frac{k}{r} (Q q_1 - q_1^2)$$

- Minimize U w.r.t q_1 : $\frac{dU}{dq_1} = k/r(Q - 2q_1) = 0 \Rightarrow q_1 = Q/2$
 - Then $q_2 = Q/2$ ✔ Minimum potential energy occurs when split equally.
-

Q18. Two charges q and 4q separated by 1 m. Net force on q if third charge 2q placed at midpoint.

- Let q at x=0, 4q at x=1, 2q at x=0.5
- Force on q:
 - From 4q: $F_1 = kq \cdot 4q/1^2 = 4kq^2$ (right)
 - From 2q at midpoint: distance 0.5 $\rightarrow F_2 = kq \cdot 2q/0.5^2 = 8kq^2$ (right)
- Net force: $F = F_1 + F_2 = 12kq^2$ right

✔ Answer: $F = 12kq^2$

Q19. Three equal charges on line with equal spacing d . Find net force on middle charge.

- Charges: $q - q - q$, spacing d
 - Force on middle q :
 - Left q : $F_1 = kq^2/d^2$ right
 - Right q : $F_2 = kq^2/d^2$ left
 - Equal and opposite $\rightarrow$ Net force = 0 ☒
-

Q20. Two charges $+q$ and $-q$ separated by $2a$. A charge Q moves from infinity to midpoint. Find change in potential energy.

- Midpoint distance from each: a

$$\Delta U = kQq/a + kQ(-q)/a = kQq/a - kQq/a = 0$$

☒ Change in potential energy: $\Delta U = 0$

☒ Section 1 completed (Q1–20)

I can now move to **Section 2: Electric Field & Potential (Q21–40)** with full detailed derivations.

Do you want me to proceed with Section 2?

yes

Great! Let's move on to **Section 2: Electric Field & Potential (Q21–40)**. I'll solve each problem **step by step**, with full explanations.

SECTION 2: Electric Field & Potential (Q21–40)

Q21. Electric field at point on axial line of dipole of moment p , distance $r \gg d$: derive formula.

Solution:

- Dipole: charges $+q$ and $-q$ separated by distance d , dipole moment $p = qd$
- Point on **axial line** (along dipole axis), distance $r \gg d$
- Electric field due to $+q$ at distance $r - d/2 \approx r$:

$$E_+ = k \frac{q}{(r - d/2)^2} \approx k \frac{q}{r^2} \left(1 + \frac{d}{r}\right)$$

- Electric field due to $-q$ at distance $r + d/2 \approx r$:

$$E_- = k \frac{q}{(r + d/2)^2} \approx k \frac{q}{r^2} \left(1 - \frac{d}{r}\right)$$

- Net field along axis:

$$E = E_+ - E_- \approx k \frac{q}{r^2} \left(1 + \frac{d}{r} - \left(1 - \frac{d}{r}\right)\right) = k \frac{2qd}{r^3} = \frac{2kp}{r^3}$$

✓ Answer: $E_{\text{axial}} = \frac{2kp}{r^3}$

Q22. Electric field at equatorial line of dipole. Compare with axial field.

Solution:

- Point on **equatorial plane** (perpendicular bisector)
- Distance from each charge $\approx r \gg d$
- Field magnitude along line connecting charges cancels; net field **toward axis**:

$$E_{\text{eq}} = \frac{kqd}{r^3} = \frac{kp}{r^3}$$

✓ Comparison: $E_{\text{axial}} = 2E_{\text{equatorial}}$

Q23. Potential at point on axis of dipole: derive expression.

- Potential is **scalar**, add contributions:

$$V = k \frac{q}{r - d/2} - k \frac{q}{r + d/2} \approx kq \left(\frac{1}{r - d/2} - \frac{1}{r + d/2} \right) \approx kq \frac{d}{r^2} = \frac{kp}{r^2}$$

✓ Answer: $V_{\text{axial}} = \frac{kp}{r^2}$

Q24. Potential at point on equatorial plane of dipole.

- Contributions from +q and -q:

$$V = k\frac{q}{r} + k\frac{-q}{r} = 0$$

✓ Answer: $V_{\text{equatorial}} = 0$

Q25. Uniformly charged rod of length L , linear charge density λ . Find electric field at point on axis at distance x from one end.

- Take rod along x-axis, end at origin to $x = L$, point at $x = x + L$
- Element: $dq = \lambda dx'$ at position x'
- Field along axis: $dE = kdq/r^2 = k\lambda dx'/(x + x')^2$

$$E = k\lambda \int_0^L \frac{dx'}{(x + x')^2} = k\lambda \left[-\frac{1}{x + x'} \right]_0^L = k\lambda \left(\frac{1}{x} - \frac{1}{x + L} \right)$$

✓ Answer: $E = k\lambda \left(\frac{1}{x} - \frac{1}{x+L} \right)$

Q26. Uniformly charged ring, radius R , total charge Q . Find electric field at point on axis.

- Ring in xy-plane, point on z-axis at distance z
- Element dq produces $dE_z = kdqz/(z^2 + R^2)^{3/2}$

$$E = \frac{kz}{(z^2 + R^2)^{3/2}} \int dq = \frac{kQz}{(z^2 + R^2)^{3/2}}$$

✓ Answer: $E = \frac{kQz}{(z^2 + R^2)^{3/2}}$

Q27. Uniformly charged ring. Find potential at axial point.

- Scalar sum: $V = k \sum dq/r = k \int dq/\sqrt{R^2 + z^2} = kQ/\sqrt{R^2 + z^2}$

✓ Answer: $V = \frac{kQ}{\sqrt{R^2 + z^2}}$

Q28. Electric field at center of square due to four charges at corners.

- Square side a , charges q_1, q_2, q_3, q_4
- Field due to charge at corner: $E = kq/r^2$, distance to center $r = a/\sqrt{2}$

- Vector sum depends on arrangement:
 - If all equal $+q$, symmetric $\rightarrow$ forces **cancel**

$$E_{\text{net}} = 0$$

Q29. Electric potential at center of square due to four charges.

- Scalar sum:

$$V = \sum \frac{kq_i}{r_i} = 4 \cdot \frac{kq}{a/\sqrt{2}} = \frac{4\sqrt{2}kq}{a}$$

✓ Answer: $V_{\text{center}} = \frac{4\sqrt{2}kq}{a}$

Q30. Electric field due to uniformly charged disk at axial point.

- Disk radius R , surface charge density σ , distance z along axis

$$dE = k \frac{dq \cdot z}{(r^2 + z^2)^{3/2}}, \quad dq = \sigma 2\pi r dr$$

$$E = 2\pi k \sigma z \int_0^R \frac{r dr}{(r^2 + z^2)^{3/2}} = 2\pi k \sigma \left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right)$$

✓ Answer: $E = 2\pi k \sigma \left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right)$

Q31. Electric potential due to uniformly charged disk at axial point.

$$V = k \int \frac{dq}{r} = k \int_0^R \frac{\sigma 2\pi r dr}{\sqrt{r^2 + z^2}} = 2\pi k \sigma (\sqrt{R^2 + z^2} - z)$$

✓ Answer: $V = 2\pi k \sigma (\sqrt{R^2 + z^2} - z)$

Q32. Electric field at distance r inside uniformly charged solid sphere.

- Sphere charge Q , radius R , $r < R$
- Use Gauss's law:

$$E \cdot 4\pi r^2 = \frac{Q_{\text{enclosed}}}{\epsilon_0}, \quad Q_{\text{enclosed}} = Q(r^3/R^3)$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3} = k \frac{Qr}{R^3}$$

✓ Answer: $E = kQr/R^3$

Q33. Electric field at distance $r > R$ outside uniformly charged sphere.

- Treat as point charge:

$$E = k \frac{Q}{r^2}$$

Q34. Potential at surface of uniformly charged sphere.

$$V = k \frac{Q}{R}$$

Q35. Potential inside uniformly charged sphere.

- From Gauss's law, $E = kQr/R^3$, potential from infinity:

$$V(r) = kQ \left(\frac{3R^2 - r^2}{2R^3} \right)$$

✓ Answer: $V(r) = \frac{kQ}{2R} (3 - r^2/R^2)$

Q36. Work done in moving a charge along equipotential surface.

- By definition, $\vec{F} \cdot d\vec{s} = q\vec{E} \cdot d\vec{s} = 0$

$$\boxed{W = 0}$$

Q37. Equipotential surfaces of a dipole: sketch and explain.

- Equations from $V = kp \cos \theta / r^2 = \text{constant}$

$$r = \sqrt{\frac{kp \cos \theta}{V_0}}$$

- Surfaces are curved lines along dipole axis, concentric along axis.
-

Q38. Two dipoles along same axis. Find net electric field at midpoint.

- Each dipole: $E = 2kp/r^3$ along axis
- Use superposition along the line connecting dipoles

$$E_{\text{net}} = 2kp_1/r_1^3 \pm 2kp_2/r_2^3$$

Q39. Charge moves along field line. Find relation between work and potential difference.

$$W = q\Delta V$$

Q40. Two charges +q and -q separated by 4 cm. Find electric field at midpoint and potential.

- Midpoint distance = 2 cm from each
 - Electric field: $E = kq/r^2 + kq/r^2 = 2kq/(0.02)^2$ along -q direction
 - Potential: $V = kq/r - kq/r = 0$
-

✔ Section 2 completed (Q21–40)

We can proceed to **Section 3: Electric Flux & Gauss's Law (Q41–60)** next.

Do you want me to continue with Section 3?

yes

Perfect! Let's move on to **Section 3: Electric Flux & Gauss's Law (Q41–60)** with detailed step-by-step solutions.

SECTION 3: Electric Flux & Gauss's Law (Q41–60)

Q41. A uniformly charged infinite plane sheet. Use Gauss's law to derive electric field.

- Consider plane with surface charge density σ
- Symmetry $\rightarrow \vec{E}$ perpendicular to plane, same on both sides
- Gaussian surface: cylinder perpendicular to plane

$$\Phi = E \cdot 2A = \frac{q_{\text{enclosed}}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0} \Rightarrow E = \frac{\sigma}{2\epsilon_0}$$

✓ Answer: $E = \frac{\sigma}{2\epsilon_0}$, uniform, independent of distance.

Q42. Electric flux through cube enclosing charge Q .

- Gauss's law: $\Phi = \frac{Q}{\epsilon_0}$
- Flux through **entire closed surface** is independent of shape $\rightarrow$ cube same as sphere

✓ Answer: $\Phi = Q/\epsilon_0$

Q43. Flux through one face of cube.

- Cube has 6 faces, charge at center

$$\Phi_{\text{face}} = \frac{\Phi_{\text{total}}}{6} = \frac{Q}{6\epsilon_0}$$

Q44. Hollow spherical shell of radius R , charge Q . Find electric field inside and outside.

- Inside: $r < R$, Gaussian surface inside $\rightarrow q_{\text{enclosed}} = 0$

$$E_{\text{inside}} = 0$$

- Outside: $r > R$, treat as point charge

$$E_{\text{outside}} = k \frac{Q}{r^2}$$

Q45. Non-conducting solid sphere, uniform charge density ρ . Find E at distance $r < R$.

$$Q_{\text{enclosed}} = \frac{4}{3}\pi r^3 \rho, \quad E \cdot 4\pi r^2 = \frac{Q_{\text{enclosed}}}{\epsilon_0} \Rightarrow E = \frac{\rho r}{3\epsilon_0}$$

Q46. Electric field of infinite line charge using Gauss's law.

- Line charge density λ
- Gaussian surface: cylinder radius r , length L

$$E \cdot 2\pi rL = \frac{\lambda L}{\epsilon_0} \Rightarrow E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Q47. Coaxial cylindrical conductor, inner radius a , outer radius b . Find E between conductors.

- Inner cylinder: charge $+Q$
- Gaussian cylinder radius r , $a < r < b$

$$E \cdot 2\pi rL = \frac{Q}{\epsilon_0} \Rightarrow E = \frac{Q}{2\pi\epsilon_0 rL}$$

Q48. Flux through cylindrical surface enclosing line charge λ .

$$\Phi = E \cdot 2\pi rL = \frac{\lambda L}{\epsilon_0} \Rightarrow \Phi = \frac{\lambda L}{\epsilon_0}$$

Q49. Hollow conductor with cavity. Point charge inside cavity. Find field inside cavity and outside conductor.

- Inside cavity: field same as point charge + induced charges on cavity surface
 - Outside conductor: total enclosed charge = 0 $\rightarrow E_{\text{outside}} = 0$
-

Q50. Electric field just outside surface of charged conductor.

$$E = \frac{\sigma}{\epsilon_0}, \quad \text{normal to surface}$$

Q51. Two coaxial spherical shells, inner Q , outer uncharged. Find field in different regions.

- Region 1: $r < R_1 \rightarrow E = 0$
- Region 2: $R_1 < r < R_2 \rightarrow E = kQ/r^2$

- Region 3: $r > R_2 \rightarrow E = kQ/r^2$
-

Q52. Field inside conductor in electrostatic equilibrium.

✓ $E = 0$

Q53. Potential inside conductor.

- Uniform, same as surface

$$V = \text{constant}$$

Q54. Flux through open surface in uniform field.

- Open surface: $\Phi = \vec{E} \cdot \vec{A} = EA \cos \theta$
-

Q55. Non-uniform charge distribution in sphere: $\rho \propto r$. Find E using Gauss's law.

- $\rho = kr$, total charge enclosed at radius r

$$Q_{\text{enclosed}} = \int_0^r 4\pi r'^2 \rho dr' = 4\pi k \int_0^r r'^3 dr' = \pi k r^4$$

$$E \cdot 4\pi r^2 = \frac{\pi k r^4}{\epsilon_0} \Rightarrow E = \frac{k r^2}{4\epsilon_0}$$

Q56. Thin spherical shell, surface charge density $\sigma(\theta) = \sigma_0 \cos \theta$. Find approximate field at pole.

- Approximation: Treat patch near pole

$$E \approx \frac{\sigma_0}{2\epsilon_0}$$

Q57. Field at edge of finite sheet.

- Non-uniform, needs integration $\rightarrow$ approximately half of central field:

$$E \sim \frac{\sigma}{2\epsilon_0} \left(1 - \frac{1}{\sqrt{2}}\right) \text{ depending on geometry}$$

Q58. Flux through cube with one vertex at charge Q .

- Flux shared equally among 8 cubes meeting at vertex

$$\Phi = \frac{1}{8} \frac{Q}{\epsilon_0} = \frac{Q}{8\epsilon_0}$$

Q59. Electric field due to charged disk. Compare with field due to plane sheet.

- Disk: $E = 2\pi k\sigma(1 - \frac{z}{\sqrt{z^2 + R^2}})$
 - Infinite plane: $E = \sigma/2\epsilon_0 = 2\pi k\sigma$ (limit $R \rightarrow \infty$)
-

Q60. Derive expression for field of spherical shell using integration (not only Gauss's law).

- Consider ring element at angle θ
 - $dE = (1/4\pi\epsilon_0) \cdot dq \cdot \cos \theta / r^2$
 - Integrate over $\theta = 0$ to π
 - Result:
 - Outside $r > R \rightarrow E = kQ/r^2$
 - Inside $r < R \rightarrow E = 0$
-

✅ **Section 3 completed (Q41–60)**

We can proceed next to **Section 4: Capacitance, Energy, & Dielectrics (Q61–90)**.

Do you want me to continue with Section 4?

yes in detailed expanded form

Perfect! Let's dive into **Section 4: Capacitance, Energy, & Dielectrics (Q61–90)** in **fully detailed, step-by-step explanations**. I'll derive formulas carefully and explain reasoning.

SECTION 4: Capacitance, Energy, & Dielectrics (Q61–90)

Q61. Parallel plate capacitor, area A , separation d , dielectric K . Derive expression for C .

- Capacitance: $C = Q/V$
- Electric field in dielectric: $E = V/d$
- With dielectric: $E = \frac{\sigma}{K\epsilon_0} = \frac{Q}{K\epsilon_0 A}$

$$V = Ed = \frac{Qd}{K\epsilon_0 A} \Rightarrow C = \frac{Q}{V} = \frac{K\epsilon_0 A}{d}$$

✓ **Answer:** $C = \frac{K\epsilon_0 A}{d}$

Q62. Energy stored in capacitor at constant voltage, dielectric inserted.

- Energy formula: $U = \frac{1}{2}CV^2$
- Dielectric increases capacitance $C \rightarrow KC_0$

$$U = \frac{1}{2}(KC_0)V^2 = K \cdot \frac{1}{2}C_0V^2$$

✓ **Observation:** Energy increases because voltage is constant and charge increases.

Q63. Energy stored at constant charge.

- Charge fixed: $Q = \text{constant}$
- Energy: $U = Q^2/(2C)$
- With dielectric: $C \rightarrow KC_0$

$$U = \frac{Q^2}{2KC_0} = \frac{U_0}{K}$$

✓ **Observation:** Energy decreases because dielectric allows same charge to store at lower voltage.

Q64. Two capacitors in series, connected to battery. Find charges and voltages.

- Series formula: $\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2}$

- Voltage division: $V_1 = Q/C_1$, $V_2 = Q/C_2$, same charge Q

$$Q = C_{\text{eq}} V_{\text{battery}}, \quad V_1 = \frac{Q}{C_1}, \quad V_2 = \frac{Q}{C_2}$$

Q65. Two capacitors in parallel, connected to battery. Find energy stored.

- Parallel formula: $C_{\text{eq}} = C_1 + C_2$
- Voltage across each: V (same)

$$Q_1 = C_1 V, \quad Q_2 = C_2 V$$

$$U = \frac{1}{2} C_1 V^2 + \frac{1}{2} C_2 V^2 = \frac{1}{2} (C_1 + C_2) V^2$$

Q66. Capacitor charged, then disconnected. Dielectric inserted. Find new voltage, charge, energy.

- Initially: Q_0, V_0, C_0
- Disconnected $\rightarrow Q$ fixed
- New capacitance: $C = K C_0$
- Voltage: $V = Q/C = V_0/K$
- Energy: $U = Q^2/(2C) = U_0/K$

✓ Key point: **Voltage decreases, energy decreases, charge unchanged.**

Q67. Energy density in dielectric medium.

- Energy: $U = \frac{1}{2} C V^2$, field $E = V/d$
 - Energy density: $u = U/(\text{volume}) = \frac{1}{2} \epsilon E^2 = \frac{1}{2} K \epsilon_0 E^2$
-

Q68. Combination of capacitors in complex network. Find equivalent capacitance.

- Reduce step by step:
 - Series: $\frac{1}{C_{\text{eq}}} = \sum 1/C_i$
 - Parallel: $C_{\text{eq}} = \sum C_i$
- Example:

$$C_{\text{eq}} = (C_1 \parallel C_2) + C_3 \quad \text{etc.}$$

Q69. Capacitor initially uncharged. Connected in circuit with battery and resistances. Find final charge and energy stored.

- Final: Capacitor behaves as open circuit

$$V_C = V_{\text{battery}}, \quad Q = CV$$

- Energy: $U = \frac{1}{2}CV^2$
-

Q70. Two capacitors, one charged, one uncharged. Connected. Find final voltages, charge, energy lost.

- Initial: $Q_1 = C_1V, Q_2 = 0$
- Connected in parallel $\rightarrow$ same voltage V_f

$$V_f = \frac{Q_1}{C_1 + C_2}, \quad Q'_1 = C_1V_f, Q'_2 = C_2V_f$$

- Energy lost: $U_{\text{lost}} = U_i - U_f = \frac{1}{2}C_1V^2 - \frac{1}{2}(C_1 + C_2)V_f^2$
-

Q71. Capacitance of spherical capacitor. Derive formula.

- Spherical capacitor: inner radius R_1 , outer radius R_2

$$V = \int_{R_1}^{R_2} E dr = \int_{R_1}^{R_2} \frac{Q}{4\pi\epsilon_0 r^2} dr = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$
$$C = \frac{Q}{V} = 4\pi\epsilon_0 \frac{R_1 R_2}{R_2 - R_1}$$

Q72. Coaxial cylindrical capacitor. Derive formula.

- Inner radius a , outer radius b , length L
- Electric field: $E = Q/(2\pi\epsilon_0 rL)$

$$V = \int_a^b E dr = \frac{Q}{2\pi\epsilon_0 L} \ln(b/a)$$
$$C = Q/V = \frac{2\pi\epsilon_0 L}{\ln(b/a)}$$

Q73. Capacitor with partially inserted dielectric. Find force on dielectric slab.

- Energy method: $F = \frac{dU}{dx}$

$$U = \frac{1}{2}C(x)V^2 \Rightarrow F = \frac{1}{2}V^2 \frac{dC}{dx}$$

Q74. Parallel plate capacitor with slab at angle. Find torque on slab.

- Torque: $\tau = \frac{dU}{d\theta} \sim \frac{1}{2}V^2 \frac{dC}{d\theta}$
-

Q75. Two identical charged capacitors connected together. Energy loss calculation.

- Initial: $U_i = 2 \cdot \frac{1}{2}CV^2 = CV^2$
 - After connection (parallel): $Q_{\text{total}} = 2CV$, $V_f = Q_{\text{total}}/(2C) = V$
 - Check: Energy decreases due to redistribution, energy lost = Joule heating
-

Q76. Energy density in parallel plate capacitor in terms of E and ϵ .

$$u = \frac{1}{2}\epsilon E^2$$

Q77. Electric field distribution inside capacitor with layered dielectrics.

- Series combination: $V = V_1 + V_2$

$$E_1 = \frac{V_1}{d_1} = \frac{\sigma}{\epsilon_1}, \quad E_2 = \frac{V_2}{d_2} = \frac{\sigma}{\epsilon_2}$$

Q78. Voltage division in series combination.

- Series capacitors: same charge Q

$$V_i = Q/C_i, \quad V_{\text{total}} = \sum V_i$$

Q79. Charge division in parallel combination.

- Parallel: same voltage V

$$Q_i = C_i V$$

Q80. Time constant for RC charging of capacitor.

$$\tau = RC$$

- Charging equation: $q(t) = Q_{\max}(1 - e^{-t/RC})$
-

Q81. Energy stored in capacitor after charging and discharging cycle.

- Maximum: $U_{\max} = \frac{1}{2} C V^2$
 - During discharging, energy dissipated in resistor
-

Q82. Work done in moving charge in capacitor system.

$$W = \Delta U = \frac{1}{2} C (V_{\text{final}}^2 - V_{\text{initial}}^2)$$

Q83. Field inside dielectric slab partially inserted in charged capacitor.

- Field inside slab: $E = V/d_{\text{effective}} = \sigma/(\epsilon_0 K)$
-

Q84. Derive expression for torque on dipole between capacitor plates.

- Energy: $U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$

$$\tau = -\frac{dU}{d\theta} = pE \sin \theta$$

Q85. Capacitance of non-uniform plates.

- Integrate over surface: $C = \int dC = \int \epsilon_0 dA/d(x, y)$
-

Q86. Series-parallel network: compute total energy.

- Reduce network $\rightarrow$ find equivalent C_{eq}
 - Energy: $U = \frac{1}{2} C_{\text{eq}} V^2$
-

Q87. Dipole near charged plate. Find force and torque.

- Force: $F = (\vec{p} \cdot \nabla) \vec{E}$
 - Torque: $\tau = \vec{p} \times \vec{E}$
-

Q88. Capacitor network: find maximum energy stored in one capacitor.

- Use charge division formulas $\rightarrow U = \frac{1}{2} CV^2$
-

Q89. Charge redistribution problem with dielectrics and multiple capacitors.

- Step 1: Find initial charges/voltages
 - Step 2: Connect capacitors $\rightarrow$ find final voltages using Q conservation
 - Step 3: Energy change $\rightarrow$ calculate energy lost
-

Q90. Energy stored in combination with battery disconnected.

- Energy: $U = \frac{1}{2} \sum_i \frac{Q_i^2}{C_i}$, sum over all capacitors
-

✅ Section 4 completed (Q61–90)

We can now move to **Section 5: Dipoles & Advanced Problems (Q91–120)**.

Do you want me to continue with Section 5 in the same detailed, expanded format?

yes

Perfect! Let's go through **Section 5: Dipoles & Advanced Problems (Q91–120)** in **fully detailed, step-by-step explanations**. I'll carefully derive all formulas and explain reasoning.

SECTION 5: Dipoles & Advanced Problems (Q91–120)

Q91. Dipole of moment $\vec{p}$ in uniform $\vec{E}$. Find torque, potential energy at angle θ .

- Torque on dipole: $\vec{\tau} = \vec{p} \times \vec{E}$

$$\tau = pE \sin \theta$$

- Potential energy of dipole: $U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$
-

Q92. Dipole along axial line of another dipole. Find net field at midpoint.

- Axial field of dipole at distance r : $E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$
- Midpoint between two identical dipoles along same axis:

$$E_{\text{net}} = E_1 - E_2 \quad (\text{consider directions})$$

Q93. Dipole in non-uniform field. Derive expression for force.

- Force on dipole: $\vec{F} = (\vec{p} \cdot \nabla) \vec{E}$
- Component-wise:

$$F_x = p_x \frac{\partial E_x}{\partial x} + p_y \frac{\partial E_x}{\partial y} + p_z \frac{\partial E_x}{\partial z}, \text{ etc.}$$

Q94. Two dipoles, perpendicular axes. Find torque and potential energy.

- Torque: $\vec{\tau} = \vec{p}_1 \times \vec{E}_2$
 - Energy: $U = -\vec{p}_1 \cdot \vec{E}_2$
-

Q95. Equipotential surfaces of dipole: sketch, derive equation.

- Potential of dipole: $V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$
 - Constant $V \rightarrow r^2 / \cos \theta = \text{constant}$
 - Surfaces: distorted spheres along dipole axis
-

Q96. Field at point on perpendicular bisector of dipole.

- Equatorial line: $r \gg d$

$$E_{\text{equatorial}} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3}, \text{ perpendicular to axis}$$

Q97. Charge moves along equatorial line of dipole. Find work done.

- Work = potential energy difference:

$$W = q(V_f - V_i)$$

- Equatorial line: $V = 0$, so if starting/ending on equatorial line far away $\rightarrow W = 0$
-

Q98. Dipole near charged plane. Find torque and energy.

- Electric field of plane: $E = \sigma/2\epsilon_0$
 - Torque: $\tau = pE \sin \theta$
 - Potential energy: $U = -pE \cos \theta$
-

Q99. Electric field at centroid of charged polygon.

- Use **superposition** of all vertex charges

$$\vec{E}_{\text{centroid}} = \sum_i \frac{kq_i(\vec{r}_{\text{centroid}} - \vec{r}_i)}{|\vec{r}_{\text{centroid}} - \vec{r}_i|^3}$$

Q100. Potential energy of system of four charges at square corners.

- Charges q_1, q_2, q_3, q_4 at corners

$$U = k \sum_{i < j} \frac{q_i q_j}{r_{ij}}$$

- Side length a : 4 sides (a), 2 diagonals ($\sqrt{2}a$)
-

Q101. Two dipoles placed along same line, separated by distance. Find equilibrium configuration.

- Energy of interaction: $U = \frac{1}{4\pi\epsilon_0 r^3} [\vec{p}_1 \cdot \vec{p}_2 - 3(\vec{p}_1 \cdot \hat{r})(\vec{p}_2 \cdot \hat{r})]$
 - Minimize $U \rightarrow$ dipoles aligned anti-parallel along line of centers
-

Q102. Electric potential along axis of finite charged rod.

- Rod: length L , linear charge λ

$$V = \frac{1}{4\pi\epsilon_0} \int_0^L \frac{\lambda dx}{\sqrt{x^2 + z^2}} = \frac{\lambda}{4\pi\epsilon_0} \ln \frac{L + \sqrt{L^2 + z^2}}{z}$$

Q103. Electric field along axis of rod.

$$E_z = -\frac{dV}{dz} = \frac{\lambda}{4\pi\epsilon_0} \left[\frac{1}{z} - \frac{1}{\sqrt{L^2 + z^2}} \right]$$

Q104. Work done in assembling multiple charges in geometric configuration.

$$W = \frac{1}{2} \sum_i q_i V_i$$

- V_i = potential at i due to all previously placed charges
-

Q105. Potential energy of continuous charge distribution.

- Continuous form:

$$U = \frac{1}{2} \int \rho(\vec{r}) V(\vec{r}) d\tau$$

Q106. Energy density in field of point charge.

$$u = \frac{\epsilon_0 E^2}{2} = \frac{\epsilon_0}{2} \left(\frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \right)^2 = \frac{Q^2}{32\pi^2\epsilon_0 r^4}$$

Q107. Force on dipole in non-uniform field, derive using gradient.

$$\vec{F} = (\vec{p} \cdot \nabla) \vec{E}, \quad F_i = \sum_j p_j \frac{\partial E_i}{\partial x_j}$$

Q108. Capacitance of composite dielectric system.

- Series: $1/C_{eq} = \sum_i d_i/(\epsilon_i A)$

- Parallel: $C_{\text{eq}} = \sum_i \epsilon_i A_i / d_i$
-

Q109. Charge distribution on isolated conductor.

- Electrostatics → **charge resides on surface**, field inside conductor = 0
-

Q110. Electric field near surface with surface charge density $\sigma(x, y)$.

$$E = \frac{\sigma(x, y)}{\epsilon_0} \hat{n} \quad (\text{perpendicular to surface})$$

Q111. Parallel plate capacitor with non-uniform separation. Find C .

- Approximate as series of strips:

$$C = \int \frac{\epsilon_0 dA}{d(x, y)}$$

Q112. Two capacitors connected: find voltage and charge redistribution.

- Use charge conservation and same voltage for parallel / same charge for series.
-

Q113. Energy considerations in moving dielectric slab.

- Energy method: $F = dU/dx$, energy decreases as slab enters capacitor
-

Q114. Torque on dipole at arbitrary orientation in non-uniform field.

- General torque: $\vec{\tau} = \vec{p} \times \vec{E}$
 - Energy: $U = -\vec{p} \cdot \vec{E}$
-

Q115. Multiple charges forming dipole array: compute net potential energy.

$$U = k \sum_{i < j} \frac{q_i q_j}{r_{ij}}$$

Q116. Capacitance of combination of spherical and cylindrical capacitors.

- Reduce step by step:
 - Spherical: $C_s = 4\pi\epsilon_0 \frac{R_1 R_2}{R_2 - R_1}$
 - Cylindrical: $C_c = 2\pi\epsilon_0 L / \ln(b/a)$
 - Combine in series/parallel
-

Q117. Energy stored in complex RC network.

- Compute equivalent C_{eq}

$$U = \frac{1}{2} C_{eq} V^2$$

Q118. Electric field along axis of charged ring.

- Ring: radius R , charge Q , point at distance x along axis

$$E = \frac{kQx}{(x^2 + R^2)^{3/2}}$$

Q119. Electric field at arbitrary point due to system of charges.

- Vector sum: $\vec{E} = \sum_i k \frac{q_i(\vec{r} - \vec{r}_i)}{|\vec{r} - \vec{r}_i|^3}$
-

Q120. Derive general expression for energy stored in N-point charges.

$$U = k \sum_{i < j} \frac{q_i q_j}{r_{ij}}$$